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Optimising the CO_2 Storage Potential in Deep Saline Aquifer by Creating Pressure Sink

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Abstract

Malaysia has pledged to reduce the intensity of Green House Gas (GHG) emission by 45% based on Gross Production Product (GDP) in 2030 and ultimately net-zero carbon emission in 2050 during UNFCCC-COP 26 in Scotland. However, the growing energy demands need some of the high contaminant (CO_2) natural gas fields in the region to be developed. To mitigate the negative climatic impact of such developments and manage the huge quantities of produced CO_2, Carbon Capture and Storage (CCS) is the only viable technology in the current world. In this paper, a multidisciplinary subsurface study is conducted to unlock the potential containment for CO_2 storage and estimate the effective volume in deep saline aquifer.

A fit for purpose 3D geological model was built for K reservoir in Z field. A range of porosity was tabulated from well log data and permeability was estimated from available poro-perm cross plot. Then, a compositional dynamic model with appropriate CO_2 storage features is established using CMG-GEM simulator. The initial formation pressure and temperature were estimated from available well tests, and PVT and SCAL data are considered from an analogue field. The base case model was initialized, and several scenarios were run to estimate the optimum effective CO_2 storage capacity for injection period of 25 years.

The CO_2 gas injection started on the January 2025 with 60 MMscf/d injection rate and maximum water withdrawal was set at 30 kbbl/d. The prediction ran until 2050. An effective CO_2 storage capacity for non-voidage case is estimated at 4.9 Bscf for a commingled injector with no producer case. This contributed only 3% of the theoretical storage capacity that can be sequestered due to high pressure build-up in the reservoir. For voidage case, one injector and one producer (commingled) case predicted to effectively sequestered 80.13 Bscf, which is approximately 51% of the total theoretical storage capacity. The simulation was further optimized with two injector and one producer (commingled) case where a total of 95.44 Bscf gas is predicted to be sequestered in the reservoir. This contributed to the highest effective CO_2 storage volume, approximately at 61% of the total theoretical storage capacity.

This study presents the feasibility and optimization of CO_2 sequestration in saline aquifer zone for one of Malaysia pre-development fields that can be as part of forecasting the CO_2 injectivity and behaviour in storage reservoir during containment. Based on the study outcomes, it was well understood that the small, restricted saline aquifers may not present the greatest avenue for sustainable and economic CO_2

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sequestration due to their limited size and the requirement of water production to create additional voidage/ pressure sink. In such a scenario, extensive and open aquifer systems in the region may be analyzed for their containment integrity and effective storage capacities.

Introduction

Re-screening study on drilled dry structures in Malay Basin was carried-out during FY2021 as part of PETRONAS's initiatives to identify saline aquifer candidates for feasibility study and broaden the funnel of CO_2 storage sites in Peninsular Malaysia. A total of 34 candidates were ranked and shortlisted based on the evaluation of critical storage elements which are the reservoir, seal presence & effectiveness, and trap presence. This re-screening exercise was carried out utilizing the available information for saline aquifer study, well logs, exploration interpretation and other available reports. As results, six potential candidates with individual field's theoretical storage capacity ranging between 0.5-23 Tscf and four fields were identified with mid to high chance of feasibility with individual field's storage capacity ranging between ~0.05 to 4.6 Tscf. On top of that, the study also identified opportunities from X, Y and Z field with theoretical storage capacity ranging between ~1.1 to 1.8 Tscf.

The integrated subsurface study was conducted for containment assessment of X, Y and Z fields. High level containment analysis included reservoir characterization, reservoir correlation, sedimentology, structural configuration and well integrity analysis. The study concluded that two out of the three studied structures (X and Y fields) were assessed to be associated with high trap risk or no trap and hence were not analysed further for storage capacity assessment. Similarly, it was found that the J and L reservoirs in Z field were also found have no trap presence and not feasible for CO_2 storage. Only the K reservoirs in Z field fulfilled all the criteria of storage screening parameters and were selected as potential storage candidate for capacity estimation.

The exploration well, Z-1 was spudded on 17 September 2010 and reached TD at depth 3142m MDDF/ 2736 m TVDSS on 23 October 2010. It penetrated 418 m (MD) into the fractured metamorphic reservoir. Minor oil and gas were discovered in both K7.1 Sand and L1 Sand, which was the secondary objective. However, there was no hydrocarbon shows in the primary objective of the well which is the fractured metamorphic reservoir. Hydrocarbon type in K7.1 was proven by sample taken from MDT-IFA logging run. L1 Sand was interpreted as oil bearing formation from the petrophysical analysis. Static Gradient Survey (SGS) was conducted during testing and suggested that L2 Sand is a water bearing formation. MDT Dual Packer was conducted in K5 Sand and the fluid type on surface confirmed that K5 Sand is a water bearing formation.

Based on the post drill analysis (PDA), the petroleum system elements in Z field are classified as proven & effective in K & L reservoirs resulted from the HC accumulation in the reservoirs (Table 1). The potential reservoir zone for Z field Storage, J, K & L reservoir been selected. Based on analysis, reservoirs J & K selected as the best potential zones for storage in Z field whereas L reservoir been excluded due to limited net reservoir thickness in that zone (Table 2).

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Table 1—Post drill analysis (PDA) summary

Parameter	Presence	Effectiveness	Remarks
Reservoir	Proven	Proven	Z-1 well proved the presence of Early Miocene to Oligocene reservoirs, 16 – 17% porosity, ~212m gross sand, ~159m net sand thickness (Group K and L).
Source	Proven	Proven	Oil shows recorded in Group K and L. K7.1 sand proved to be oil bearing form the fluid sample taken during MDF. L1 sand interpreted as oil bearing based on the petrophysical analysis. Geochemistry study has confirmed the presence of poor to fair organic content, but the sequence is immature for oil generation and that the top of the conventional 'oil window'.
Timing/ Migration	Proven	Proven	Evidence from basin modelling suggested that reservoir predated the trap formation. The expulsion of hydrocarbon was around 10-12 Ma.
Trap/ Seal	Proven	Proven	Hydrocarbon discovery in K7.1 sand has confirmed that the overlying lacustrine deposited K Shale act as seal for this particular reservoir, whereas L1 and L2 shale are sealed by overlying L shale.

Table 2—Preliminary screening summary

Fields	Zone of interest	Depth (m)	Thickness (m)	Porosity (%)	Permeability (mD)	Pressure (psia)	Temperature (deg F)	Seal Thickness	Trap	Well Integrity
Z field	J sand					No info				
	K sand (K5 and K7.1)									
	L sand (L1 and L2)					No info				

Depth	Green	Yellow	Red
Thickness	> 1000m TVDSS	< 1000m TVDSS from seabed	
Porosity	> 50m	20-50m	< 20m
Permeability	> 20%	10-20%	< 10%
Pressure	> 100 mD	10-100 mD	< 10mD
Temperature	> 1100 psia	< 1100 psia	
Seal Thickness	> 31 deg C	< 31 deg C	
Trap	> 100m	20-100m	< 20m
Well Integrity	Presence & effective	Presence	Absence

Low Risk
Yellow
Red

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Static Modelling

Theoretical Storage Capacity Estimation

Z field target storage interval is coming from K reservoirs. K reservoirs (K7.1) are proven hydrocarbon interval based on exploration result which suggested a proven and effective reservoir. However, based on analysis, Z-1 well located on the flank of the structure whereas the crest of the structure remained untested. To assess the theoretical storage capacity in Z field, a high-level model been built to assess the storage capacity of the target intervals. The average properties from available RESSUM incorporated in the model as an input for properties parameters (NTG, Sw, Porosity). The details of each of the input parameter and grid information used in the model are summarize in Table 3 and Figure 1.

Table 3—Summary of structural grid information for Z field

Parameter	Remarks
Horizon	• As per latest geophysics interpretation
Faults	• 2 horizons, 4 faults
Well markers	Existing interpretation as per report
Model coverage (AOI)	• Covered Z field structure only • Top J to Top L shale
Properties	Average properties provided by petrophysics
Contacts	Spill point from 2 fault bounded structure
Grid dimension	100m x 100m
Total grid cells	318750
Cell geometries	No negative cells

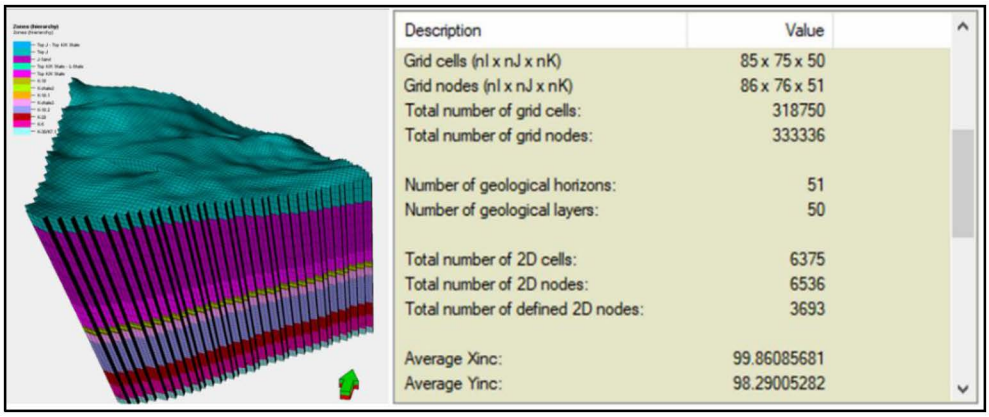


Figure 1—Z Field structural grid architecture

The theoretical storage volume has been calculated inside the designated region which represent Z field structure (Figure 2). The dummy fluid contact represent by the spill point of the Z field structure bounded by the two faults to the NE and SW is applied. The theoretical storage capacity calculated in dynamic model incorporated input from static model (grid, porosity, NTG and contact). The summary of the volumetric for K reservoir are as per Table 4. Overall, the total theoretical storage capacity for Z field is estimated at 0.16 Tscf and is classified as Low.

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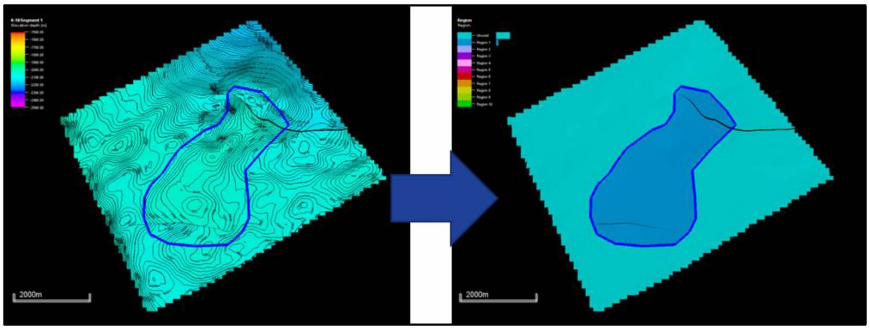


Figure 2—Region of storage capacity estimation

Table 4—Summary of Z field storage capacity in K reservoirs

Zones	Bulk volume [*10^6 m^3]	Net volume [*10^6 m^3]	Pore volume [*10^6 m^3]	Theoretical Storage Capacity [Tscf] *
K-10	50	46	5	0.04
K-10.1	32	23	2	0.01
K-10.2	840	353	2	0.01
K-20	379	310	6	0.04
K-5	410	353	5	0.03
K7.1	154	99	3	0.02
Total	**1865**	**1184**	**23**	**0.16**

Dynamic Modelling

Geomechanics Study

A geomechanical study was conducted on Z field for CO_2 geological storage evaluation. Rock properties were generated using wireline logs data. Since there is no available core data, the correlations below (Table 5) were used to convert dynamic to static rock elastic properties and to generate rock strength properties.

Table 5—Correlations used in 1D geomechanical model construction

Properties	Final Correlation Used
Young's Modulus Static (YM_STA)	John Fuller Correlation
Poisson's Ratio (PR_STA)	Multiplier: 0.9*PR_DYN
Bulk and Shear Moduli	PR_STA & YM STA as per above
Friction Angle	GR Correlation
UCS	McNally Correlation (Sandstone) Horsrud Correlation (Shale)
Tensile Strength	1/10* UCS
Failure Model	Mohr-Coulomb Criterion
Plasticity Coefficient	1.0 – 1.1
SHmax/Shmin Ratio	1.1
Poroelastic Horizontal Strain	$\epsilon H = 0.0008$ $\epsilon h = 0.0005$

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Horizontal stress, $\Delta\sigma_h$ change due to pore pressure change, ΔP_p is calculated from Poisson's ratio, υ and Biot's coefficient, α using the equation:

$$\Delta\sigma_h = \alpha\frac{1-2\nu}{1-\nu}\Delta P_p \quad \text{Equation 1}$$

Maximum vertical stress transfer ratio of 0.33 from reservoir to overburden due to injection was applied based on published numerical analysis. The recommended maximum injection pressure was determined based on caprock fracture propagation pressure with safety factor of 1.5 applied to the incremental injection pressure above the initial reservoir pressure as summarized in Table 6.

Table 6—Recommended maximum injection pressure for reservoir of interest

No	Reservoir	Reference Depth (mMD)	Initial Reservoir Pressure (psia)	Original Caprock Total Shmin (psi)	Maximum Injection Pressure (Equals Caprock Total Shmin After Injection) (psi)	Recommended Maximum Injection Pressure with 1.5 Safety Factor Applied (psi)
1	K-10	2107.35	2919.3	5904	5578	**4692**
2	K-10.1	2122.96	2940.6	5963	5610	**4720**
3	K-10.2	2158.85	2994.6	6076	5718	**4810**
4	K-20	2264.43	3145.2	6410	6042	**5076**
5	K-5	2314.39	3213.4	6575	6184	**5194**
6	K-7.1	2350.63	3288.7	6691	6294	**5292**

Effective Storage Capacity Estimation

The workflow of the modelling work applied in this project is shown in Figure 3. The modelling workflow started with 3-D static model construction from geologist. Geologist has constructed a high level 3-D model in Petrel to define the average porosity, permeability, net to gross, water saturation and other parameters utilizing available data in Z field. The average properties from available RESSUM have been incorporated into the model. The high-level static 3-D model has total number of 318,750 grid cells with 100m × 100m grid dimension and covered Z field structure only from Top J layer to Top L shale layer. In terms of volumetric, geologist has incorporated a region known as 'Region 1' to calculate the bulk volume, net volume and pore volume that represented the Z field structure for this study.

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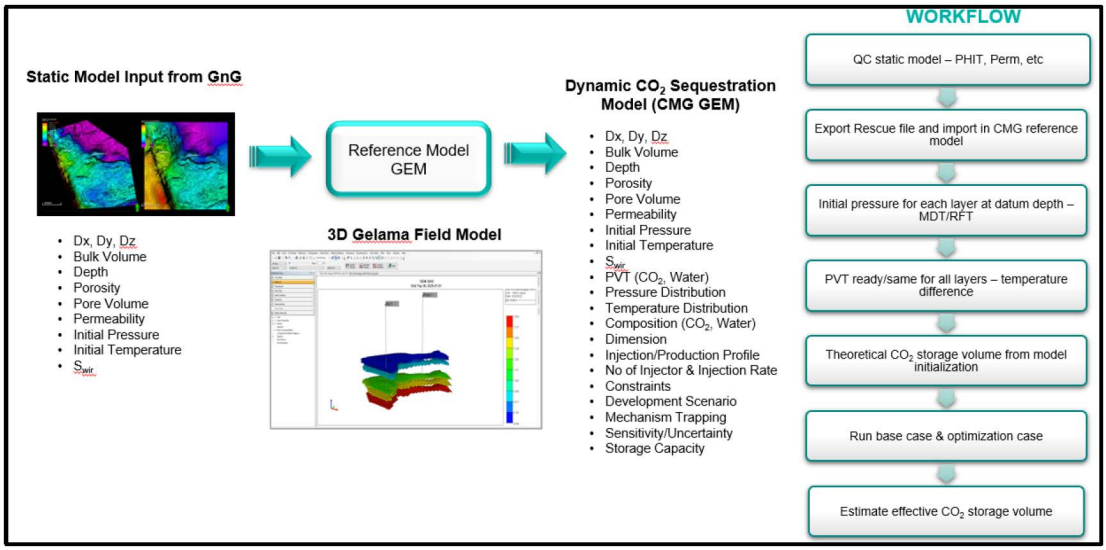


Figure 3—The storage capacity assessment workflow

The high-level static model was then passed to reservoir engineers in .bin format and exported into CMG software as reference model. In this study, CMG-GEM (version 2020.10) was used as a compositional simulator for CO_2 storage capacity assessment, injectivity and sensitivity analysis. Upon receiving the static model, the model was verified before performing the model initialization. The permeability data is estimated from nearby field. Porosity and permeability values are different for every layer in Z field but there will be only one value of porosity and permeability applied to each of the layers which are in this case, the average porosity and average permeability as input. Laterally, those layers will be like homogenous layer.

Next, the initial model condition such as datum depth, pressure at datum depth and water-gas contact in the model were defined. The formation pressure is calculated by water gradient of 0.433 psi/ft at specified mid-depth for every layers. In terms of PVT, typical PVT of gas (C_1+CO_2) is used for modelling and same PVT have been used for all layers.

After proper model initialization, the model is run with few scenarios case of commingled CO_2 injectors with and without water producer for a period of 25 years. Some constraints have been applied to CO_2 injectors such as maximum bottom hole pressure based on geomechanics analysis and maximum gas injection rate. The impact of introducing water producer is significant and allowing more spaces for CO_2 to occupy the reservoirs. The base case is further optimized to maximize the potential CO_2 storage capacity. Lastly, effective CO_2 storage volume is estimated based on cumulative CO_2 gas injection for each layer at the end of injection period and compared against theoretical storage capacity.

Modelling Input

In terms of model input parameters, the porosity, net-to-gross (NTG), Swi and contact levels are provided by G&G team. The details of information are presented in Table 7.

Table 7—Input Data and Property for Z Field Model

No	Layer	Porosity	NTG	Permeability	Contacts m	Initial pressure psi	Bg	Pressure constraint from Geomech	Initial temp, °F	Pore volume 10^6 m3	Theoretical Storage Capacity, Bcf
1	K-10	0.26	0.92	300	-2055	2919.3	0.00341	4692	193.4	5	37.01
2	K10.1	0.16	0.72	10	-2070	2940.6	0.0034	4720	194.2	2	12.24
3	K10.2	0.16	0.42	10	-2108	2994.6	0.00337	4810	198.9	2	12.35
4	K-20	0.19	0.82	50	-2214	3145.2	0.00335	5076	203.2	6	41.55
5	K-5	0.18	0.86	50	-2262	3213.4	0.00333	5194	205.6	5	34.96
6	K-7.1	0.15	0.64	10	-2315	3288.7	0.00332	5292	207.7	3	18.91
SUM										23	157.03

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The formation pressure is calculated by water gradient of 0.433 psi/ft. The typical gas-brine water permeability is used. Only Swirr is adjusted for each layer. Typical PVT of gas (C_1+CO_2) is used for modelling. The total theoretical storage capacity is estimated around 157 Bscf.

In terms of well set up in the model, it includes one injector and one producer as commingled for all target layers. The gas rate is set 60 MMscf/d with max BHP of 5000 psi considering the risk of geomechanical failure. The producer well is set with surface rate of 30,000 bbl/d and min BHP of 800 psi. the CO_2 production at the producer is also monitored and 1 MMscf/d is assumed as CO_2 breakthrough which in our case means stop of CO_2 injection and the corresponding cumulative CO_2 injection up to breakthrough time is considered as effective storage capacity. Figure 4 shows the dynamic model of Z field - K layers.

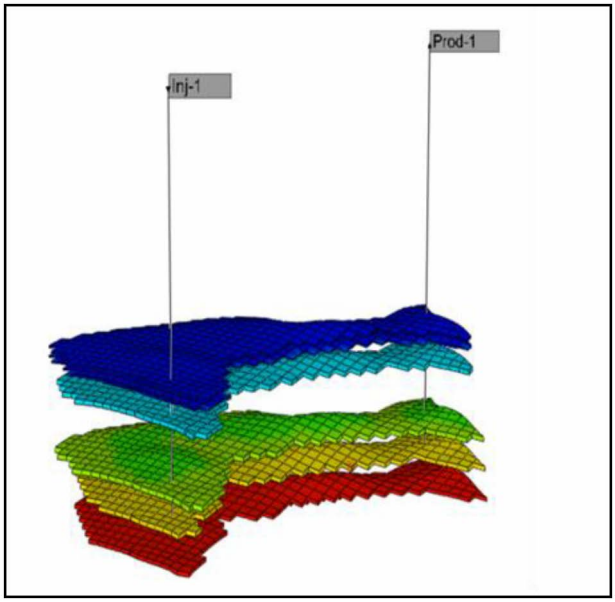


Figure 4—Dynamic model of Z field

Modelling Results

One Commingled Injector (No Producer)

One commingled injector with no produced is placed at the edge of the Z field reservoir model as shown in Figure 5 to mimic a non-voidage case model. A maximum of 60 MMscf/d of pure CO_2 gas is set to be injected and the maximum bottomhole pressure is set at 5000 psi. However, due to the initial pressure build-up in the reservoir, the injector was shut after 200 days of injection period. The effective storage gas capacity is estimated at 4.9 Bscf at the end of the simulation run. This contributes to the 3% of the theoretical storage capacity that can be sequestered. Figure 6 and Figure 7 shows the gas injection profile as well as average reservoir pressure for non-voidage case model, respectively. Clearly, for non-voidage case, the maximum geo-mechanical pressure limit has been reached. Thus, this scenario is not a suitable candidate for CO_2 storage in Z field.

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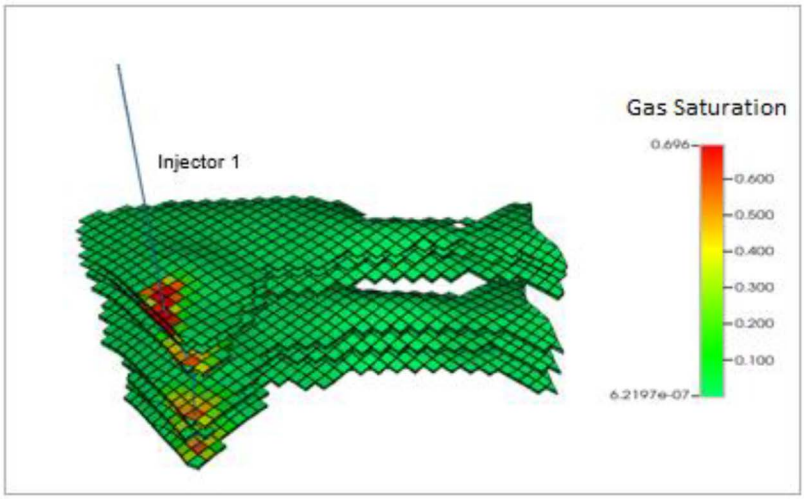


Figure 5—One commingled injector (no producer) for non-voidage case

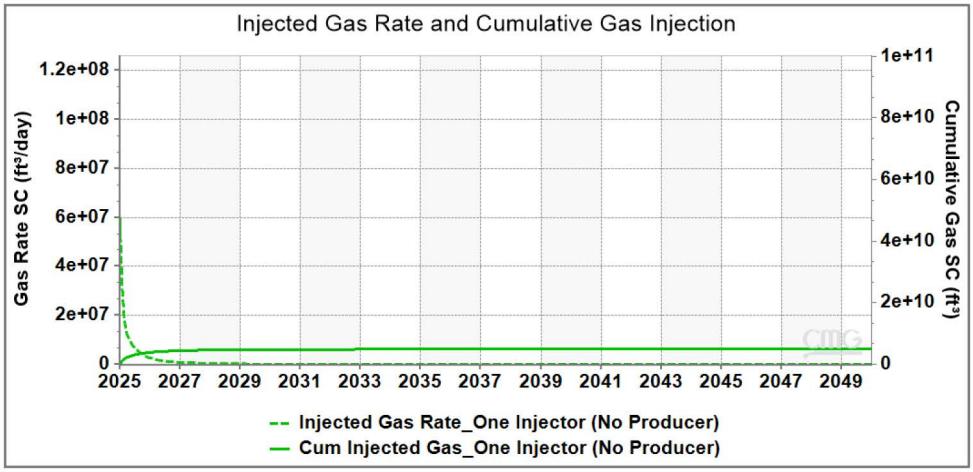


Figure 6—Gas injection profile for non-voidage case

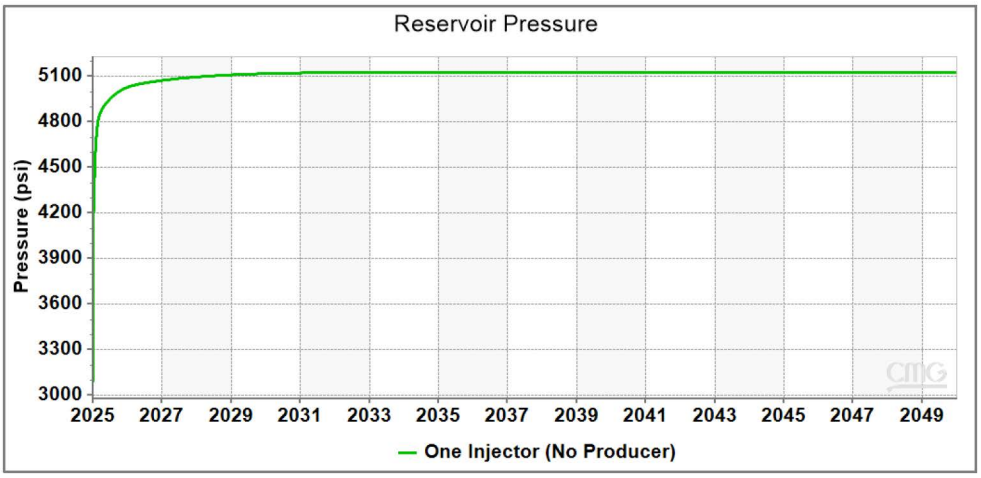


Figure 7—Average reservoir pressure for non-voidage case

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One Injector and One Producer (Commingled)

For voidage voidage case, one commingled injector with one commingled producer is set up as shown in Figure 8. The maximum gas rate injection is set at 60 MMscf/d while the maximum water production rate is set at 30k bbl/d. The minimum BHP is set at 800 psi.

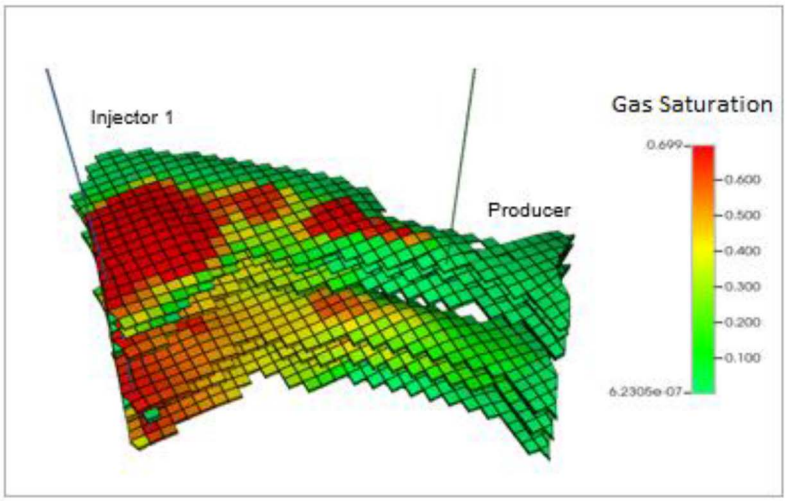


Figure 8—One Injector and One Producer (Commingled) for Voidage Case

Based on Figure 9 the effective storage capacity is estimated at 80.13 Bscf. This contributes to approximately 51% from the theoretical storage capacity. Also, three plateaus are observed in gas rate injection during the simulation period. The first plateau indicates the movement of gas in K10 zone due to a good permeability of 300 mD. The second plateau indicates the movement of injected gas in medium permeability zone which is K20 and K5 zone. While the third plateau with a lower injection rate refers to the tight permeability in K7.1 zone. The same phenomena were observed in water injection and production profile as shown in Figure 10. A good pressure trend with three steep increments was observed in Figure 11 due to the pressure build up in the reservoir during the continuous gas injection.

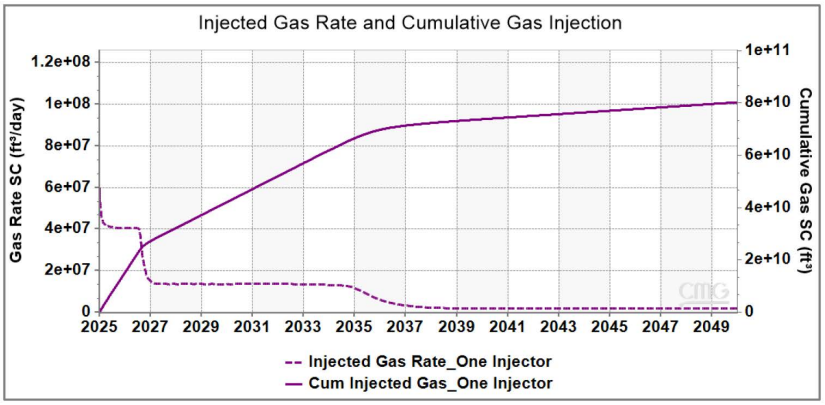


Figure 9—Gas injection profiles

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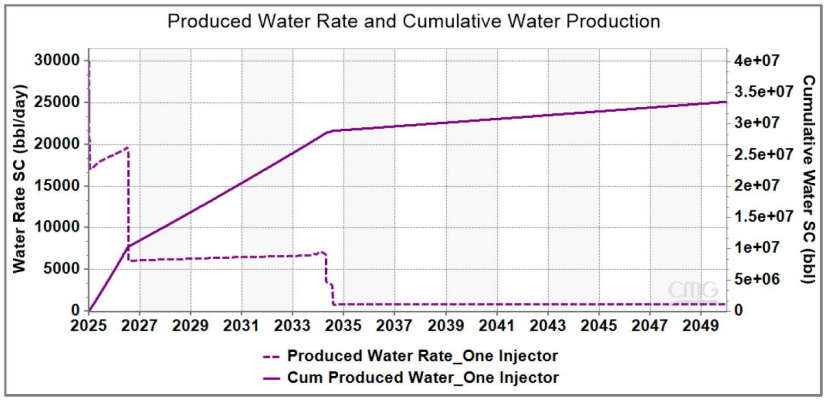


Figure 10—Water production profiles

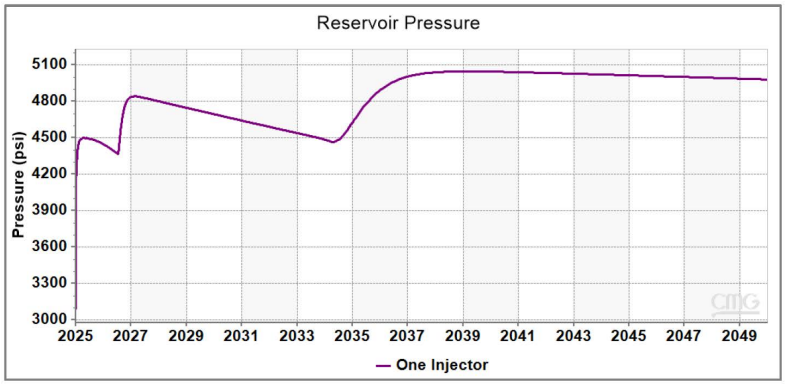


Figure 11—Average reservoir pressure

Two Injectors and One Producer (Commingled)

To further optimize on the effective CO_2 storage, two commingled injectors and one commingled producer set up as shown in Figure 12. The maximum gas injection rate and maximum water production rate were set up at 60 MMscf/d and 30k bbl/d, respectively.

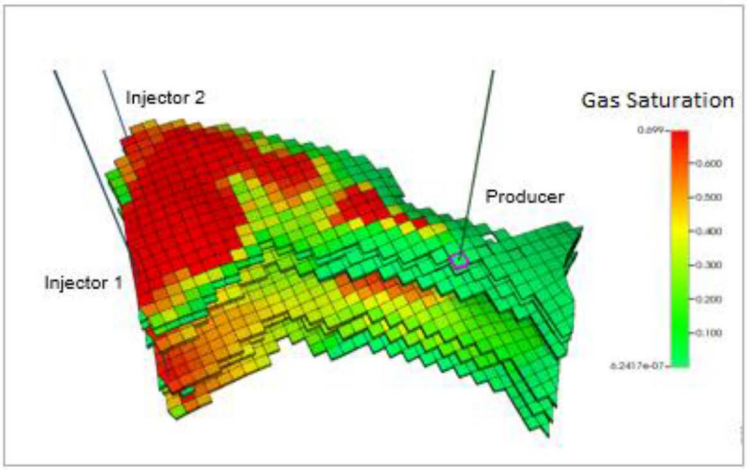


Figure 12—Two injectors and one producer (commingled) for voidage case

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Based on Figure 13, this scenario successfully predicted an effective storage capacity of 95.44 Bscf, approximately around 61% of the total theoretical CO_2 storage capacity. A total of three plateaus were observed on the water production profile in Figure 14, indicating the three different value of permeability zone in the reservoir. The total water produced from this scenario is approximately 40 Mbbl with the highest rate of 21 kbbl in the early stage of production. A good trend of average reservoir pressure is shown in Figure 15 within allowable geomechanics pressure limit.

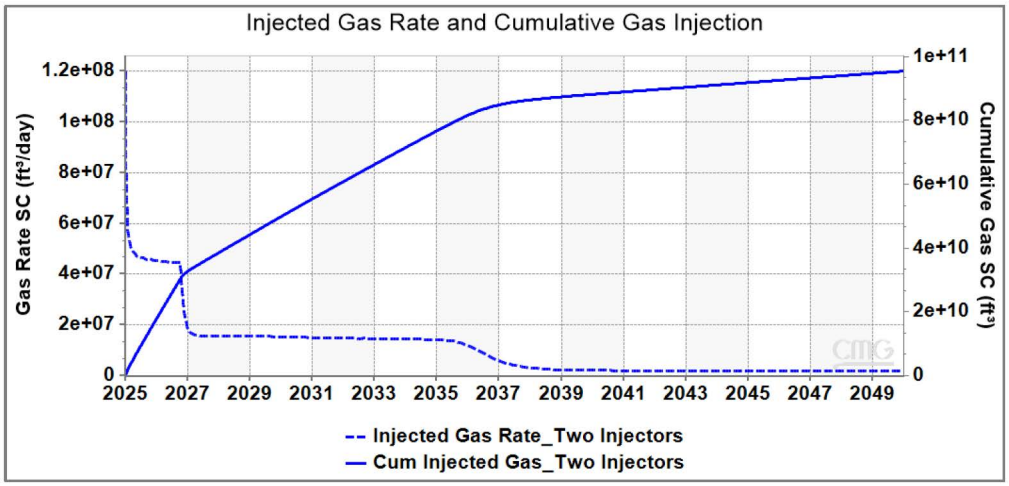


Figure 13—Gas injection profiles

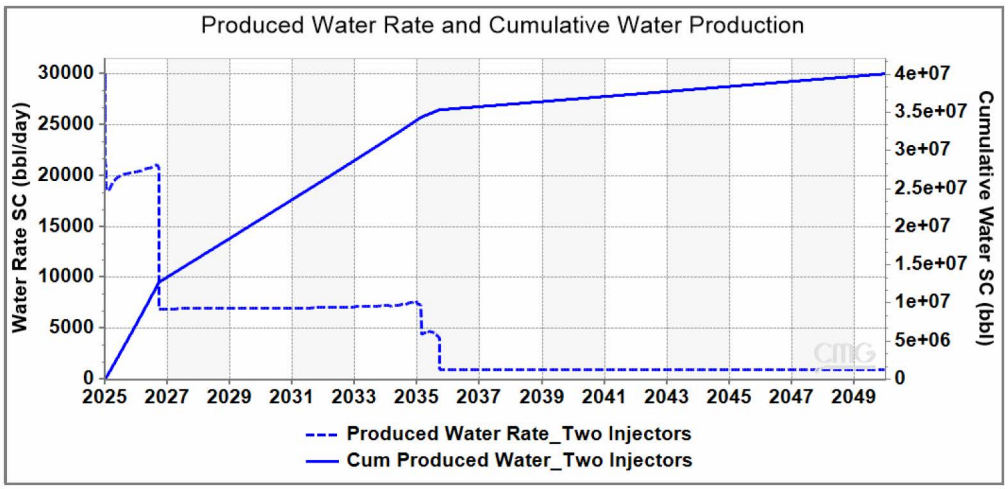


Figure 14—Water production profiles

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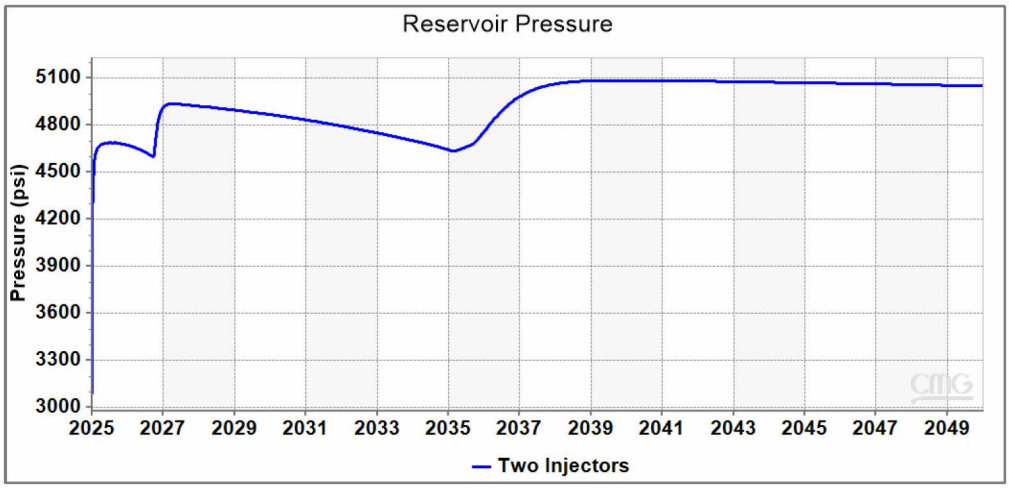


Figure 15—Average reservoir pressure

Summary

The total effective storage capacity of Z field K layers simulated in this study is shown in the table below:

Table 8—Total Effective Storage Capacity of K Layers in Z field

Field	Theoretical Storage Capacity, Bcf	Effective Storage Capacity, Bscf			% ratio (Eff / Theoretical Storage Capacity)		
		Base Case withOUT voidage	Base Case With Voidage	Optimization Case	Base Case withOUT voidage	Base Case With Voidage	Optimization Case
TOTAL	**157.03**	**4.9**	**80.13**	**95.44**	**3%**	**51%**	**61%**

In summary, the highest effective CO_2 storage volume is identified approximately 95.44 Bscf as optimized case, which is estimated around 61% of the theoretical storage volume, 157.03 Bscf. Based on this study outcomes, technically, it is not recommended to proceed further with CO_2 injection planning in Z field due to very small effective storage volume estimation and high uncertainties from the dynamc modelling inputs as of this study.

Conclusion

The dynamic simulation scenarios demonstrated the fact that the effective CO_2 storage capacity for restricted saline aquifers can be significantly enhanced by creating voidage or pressure sink by production of formation water. CCS is crucial for the timely development of high contaminant gas fields in the PM area and saline aquifers are expected to play a vital role in managing the GHG emissions associated with the development of high contaminant gas fields. However, during the current study it was well understood that the small, restricted saline aquifers may not present the greatest avenue for sustainable and economic CO_2 sequestration due to their limited size and the requirement of water production to create additional voidage/ pressure sink. In such a scenario, extensive and open aquifer systems in the region may be analyzed for their containment integrity and effective storage capacities.

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Nomenclature

UNFCCC-COP 26	United Nations Framework Convention on Climate Change – Conference of Party 26
CO_2	Carbon Dioxide
C_1	Methane
CCS	Carbon Capture, Storage
CMG-GEM	Computer Modelling Group - GEM
PVT	Pressure Volume Temperature
SCAL	Special Core Analysis
MMscf/d	Million Standard Cubic Feet per Day
Bbl/day	Barrel per Day
Bscf	Billion Standard Cubic Feet
Tscf	Trillion Standard Cubic Feet
FY2021	Financial Year 2021
PETRONAS	Petroliam Nasional Berhad
TD	Total Depth
MDDF	Measured Depth Derrick Floor
TVDSS	True Vertical Depth Sub Sea
MD	Measured Depth

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